

Linux Fundamentals Practical Report

Organization: Codomax Digital Solutions

Internship Task: Linux, Security Tools, and Reconnaissance

1. Introduction

This report summarizes my practical introduction to Linux fundamentals. I used basic terminal commands to navigate directories, create and inspect a file, change its permissions, and view user, group, and process information.

The purpose of this exercise was to become familiar with common Linux commands and understand how they support everyday system administration and security work.

2. Practical Activities

I practiced the following commands:

Command	Purpose
pwd	Displays the current working directory.
ls	Lists files and directories in the current location.
mkdir practice	Creates a directory named practice.
cd practice	Changes the current directory to practice.
touch hello.txt	Creates an empty file named hello.txt.
ls -l	Displays files with details, including permissions and ownership.
cat hello.txt	Displays the contents of hello.txt.
chmod 600 hello.txt	Changes the permissions on hello.txt.
whoami	Displays the name of the current user.
id	Displays user and group information.
ps	Displays information about running processes.

3. Learning Outcomes

This exercise helped me understand how Linux commands can be used to navigate the file system and manage files. I practiced creating a directory and a file, then used listing commands to inspect them.

I also learned that file permissions determine who can access a file and what actions they can perform. The `whoami` and `id` commands provide information about the current user and associated groups. The `ps` command displays processes that are running on the system.

I completed the listed commands, but I did not record their exact terminal output during the exercise. Therefore, this report describes the purpose of each command and the concepts practiced rather than specific observed results.

4. Practical Evidence

Figure 1. Linux command practice

```
Last login: Wed Sep 30 09:28:26 2026 from 154.192.112.74
ubuntu@ip-172-31-34-4:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-34-4:~$ ls
practical
ubuntu@ip-172-31-34-4:~$ mkdir codomax
ubuntu@ip-172-31-34-4:~$ cd codomax
ubuntu@ip-172-31-34-4:~/codomax$ touch tariq.txt
ubuntu@ip-172-31-34-4:~/codomax$ ls -la
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Sep 30 09:34 .
drwxr-x--- 6 ubuntu ubuntu 4096 Sep 30 09:34 ..
-rw-rw-r-- 1 ubuntu ubuntu    0 Sep 30 09:34 tariq.txt
ubuntu@ip-172-31-34-4:~/codomax$
```

Figure 2. File listing and permissions

```
ubuntu@ip-172-31-34-4:~/codomax$
ubuntu@ip-172-31-34-4:~/codomax$ cat tariq.txt
ubuntu@ip-172-31-34-4:~/codomax$ chmod 600 tariq.txt
ubuntu@ip-172-31-34-4:~/codomax$ ls -l tariq.txt
-rw----- 1 ubuntu ubuntu 0 Sep 30 09:34 tariq.txt
ubuntu@ip-172-31-34-4:~/codomax$
```

5. Security and Scope

This report covers basic Linux practice. It does not document active reconnaissance or testing of external systems. Any future use of security tools such as Nmap, Wireshark, Burp Suite, or OWASP ZAP should take place in an approved lab and within a clearly authorized scope.

6. Conclusion

The practical exercise provided an introduction to Linux file navigation, file creation, permissions, user and group information, and running processes. These fundamentals are useful for further study of system administration and security.